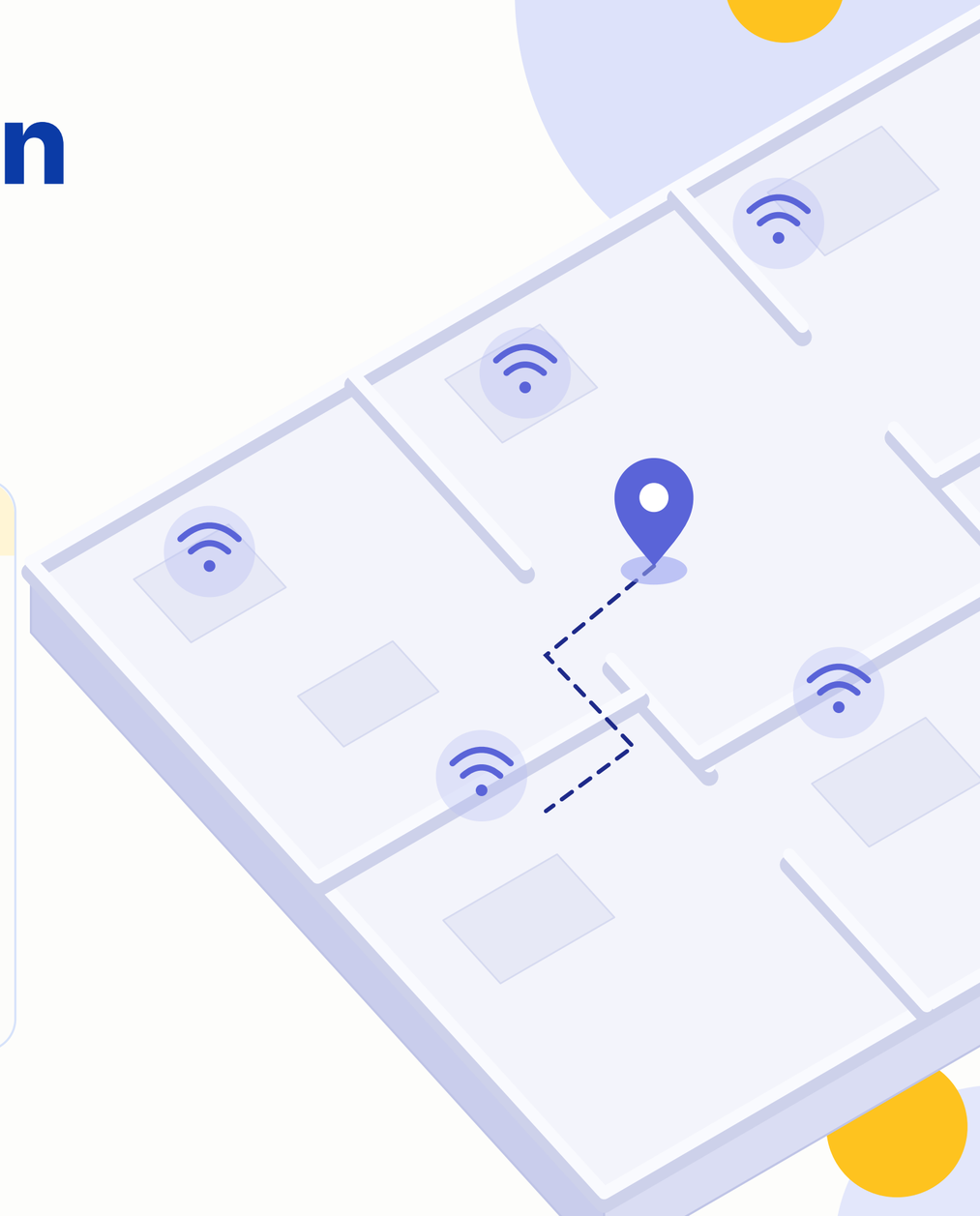
PROJECT 9

Indoor Location Prediction Using Wi-Fi Signals



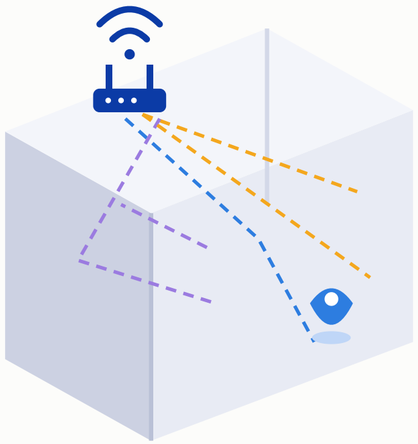
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The Problem: Why Is Indoor Positioning Difficult?

Background

- GPS signals are weakened or blocked by walls, structural steel, and other indoor obstacles.
- Indoor multipath propagation causes Wi-Fi signals to reflect, scatter, and interfere.
- Conventional geometric positioning is difficult without accurate transmitter locations and building maps.



Why Wi-Fi fingerprinting?

- Uses **existing** Wi-Fi access points without requiring additional positioning hardware.
- Matches received signal strength indicator (RSSI) measurements to learned location patterns.
- Supports indoor localization in campuses, hospitals, offices, and shopping malls.

Mathematical problem formulation

Each scan is represented by an RSSI feature vector:

$$\mathbf{x} = [r_1, r_2, \ldots, r_{520}]$$

The combined multitask objective can be expressed as:

$$L_{\text{total}} = \lambda_b L_b + \lambda_f L_f + \lambda_c L_c$$

The system performs three prediction tasks:

- **Building classification:** Predict one of 3 buildings.
- **Floor classification:** Predict one of 5 floor classes.
- **Coordinate regression:** Predict continuous (x, y) coordinates.

Key challenges: Sparse AP measurements, device and temporal variation, and the need to predict building, floor, and coordinates.

Dataset: UJIIndoorLoc Benchmark

The UJIIndoorLoc dataset, introduced by Torres-Sospedra et al. (2014), is a public benchmark for Wi-Fi-based indoor localization.

3
Campus buildings

520
Original Wi-Fi features

19,937
Original training scans

1,111
Held-out validation scans

Dataset characteristics

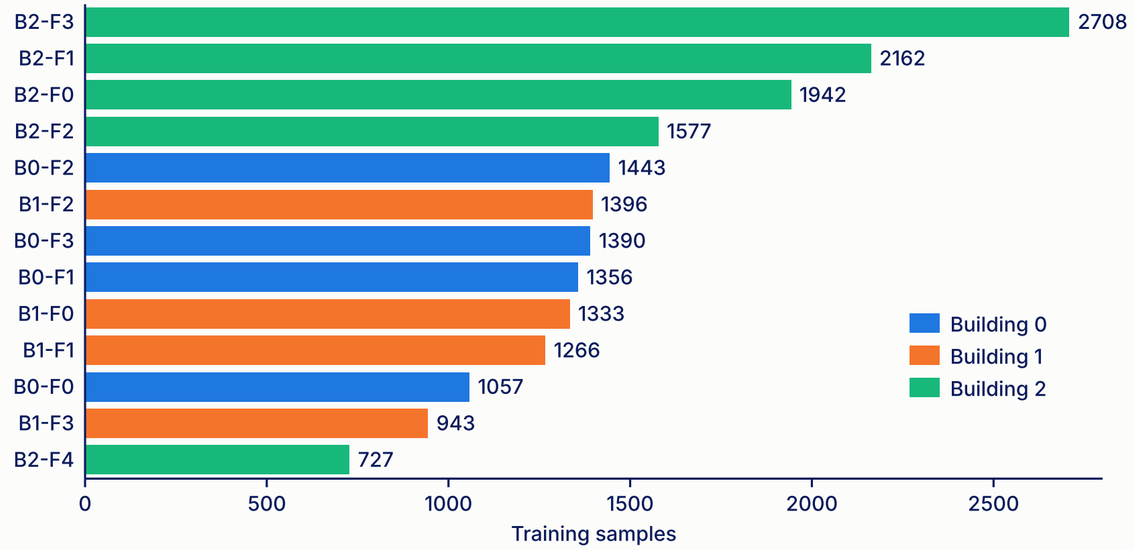
Covers approximately **110,000 m²**, with up to five floors and **13 valid building–floor combinations**.

Training data was collected in **May–June 2013**.

Validation data was collected in **September–October 2013**.

Nine of the eleven validation phones were not seen during training.

Training samples per building–floor



Why this dataset matters

It enables evaluation under **device heterogeneity** and **temporal distribution shift**, both important challenges for real-world indoor positioning.

Data Quality Audit and Leakage Prevention

Data cleaning

Issue identified	Action taken
637 duplicate scans	**Removed** duplicate records
73 empty scans	**Removed** from training
55 inactive access points	**Removed**, reducing 520 channels to **465**
775 unusually strong readings from Phone 19	**Capped** readings above −30 dBm at **−30 dBm**

Important

Random cross-validation may place scans from the same physical locations in both training and test folds. It **should not automatically be interpreted as evidence of generalization** to unseen locations.

Preprocessing Pipeline (from report)

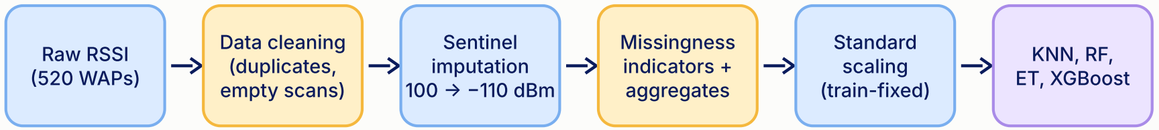


Fig. 5. Overview of the proposed preprocessing and prediction pipeline. Scaling parameters are fitted on the training set only.

Leakage prevention

- Excluded contextual metadata such as **SPACEID, RELATIVEPOSITION, USERID, PHONEID**, and **TIMESTAMP** from model inputs.
- Fitted preprocessing parameters using **training data only**.
- Used the **official held-out validation set** to assess generalization across phones and a later survey period.

Outcome: A cleaner evaluation pipeline designed to **reduce duplicate-driven performance inflation** and **metadata shortcut learning**.

Literature Review and Research Gap

Existing approaches

Study	Model	Reported findings
Torres-Sospedra et al. (2014)	1-NN	**100%** building accuracy; **~ 88%** floor accuracy
Torres-Sospedra et al. (2015)	KNN / WKNN	**~ 89–91%** floor accuracy
Salamah et al. (2016)	RF, SVM, KNN + PCA	Above **90%** zone accuracy on a private dataset
Nowicki & Wietrzykowski (2017)	MLP + stacked autoencoders	**99.8%** building; **91.3%** floor accuracy
Kim et al. (2018)	Hierarchical DNN	**99.8%** building; **91.3%** floor; 9.3 m mean error
Song et al. (2019)	CNNLoc	**~ 90%** floor accuracy; 10.2 m mean error

Reported figures may use different datasets and evaluation protocols, so they are **not directly comparable**.

Research gaps

1. Building, floor, and coordinate prediction are often trained **independently**.
2. Errors in hierarchical prediction can **propagate** to coordinate estimation.
3. Classical and deep learning models may use **different preprocessing** and evaluation protocols.
4. Accuracy can deteriorate on **unseen** phone models, users, or later survey periods.

Project objectives

- Develop a **consistent** preprocessing and feature-engineering pipeline.
- Compare **KNN, Random Forest, Extra Trees,** and **XGBoost**.
- Evaluate **building classification, floor classification,** and coordinate regression.
- Analyze **errors** across buildings, floors, devices, and validation conditions.

Central research question: Can careful Wi-Fi preprocessing and robust **classical machine learning** deliver **accurate indoor localization** under real-world distribution shifts?

Feature Engineering and Sentinel Imputation

Feature engineering pipeline

520 original AP channels → Remove **55** inactive access points → Impute missing RSSI and generate indicators

935 engineered features

465 RSSI + **465** missingness indicators + **5** aggregate statistics

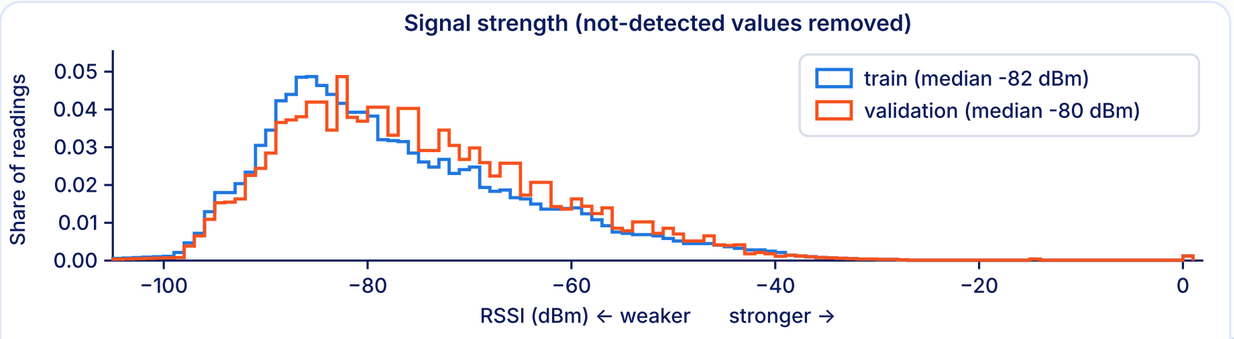
Feature groups

RSSI features	Imputed signal strength from **465** access points.
Missingness indicators	Binary values showing whether each AP was detected.
Aggregate features	AP count, maximum RSSI, mean RSSI, standard deviation, and RSSI range.
Standardization	Training-fitted **StandardScaler**, where appropriate for the model.

Sentinel imputation

The original **+100** value is a missing-data marker, not a physical RSSI measurement. It was replaced with **−110 dBm** to represent an undetected, very weak signal.

+100 → −110 dBm



Ablation study

Preprocessing configuration	Floor accuracy	Mean error
Raw sentinel	87.58%	13.83 m
Imputed RSSI	**91.27%**	**12.30 m**
Imputed + indicators	90.28%	12.30 m
Imputed + indicators + aggregates	89.74%	12.88 m
Full features + scaling	89.74%	12.81 m

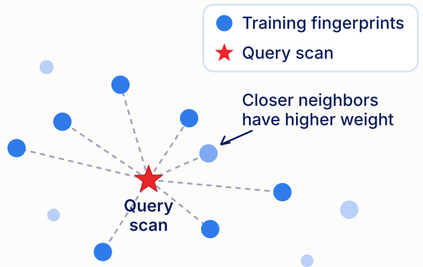
Main finding: Replacing the sentinel improved floor accuracy by **3.69 percentage points** and reduced mean positioning error by **1.53 m** in the reported experiment.

Classical Machine Learning Models

1 Distance-weighted K-Nearest Neighbors (KNN)

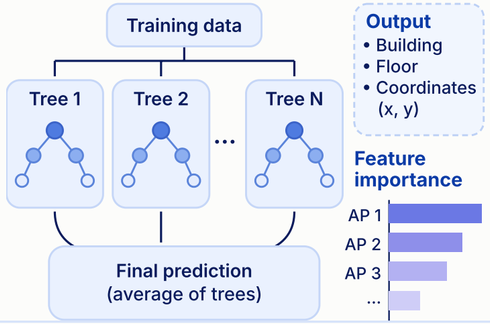
- Finds training fingerprints similar to the observed Wi-Fi scan.
- Estimates coordinates using a weighted average of neighboring locations.
- Nearby fingerprints receive greater weight.

$$\hat{y} = \frac{\Sigma_i \, w_i y_i}{\Sigma_i \, w_i}, \quad w_i = \frac{1}{d_i + \varepsilon}$$



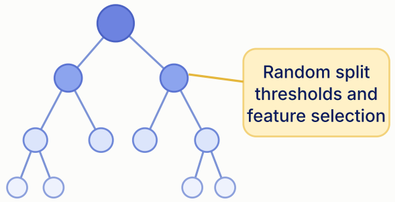
2 Random Forest (RF)

- Combines predictions from multiple decision trees.
- Uses bootstrap samples and randomized feature selection.
- Captures nonlinear patterns and provides feature-importance estimates.



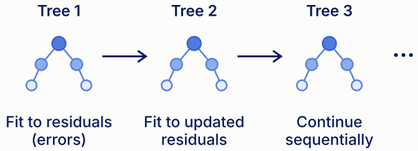
3 Extra Trees Regressor

- Introduces additional randomness into split thresholds and feature selection.
- Learns nonlinear relationships between Wi-Fi signals and coordinates.
- Used for continuous coordinate prediction.



4 XGBoost

- Builds trees sequentially to improve predictions.
- Optimizes a regularized objective using gradient and second-order information.
- Can capture complex relationships between RSSI features and location.



5 Baseline: Dummy model

- Predicts simple reference values for building, floor, and coordinates.
- Establishes baseline performance to determine whether trained models provide meaningful improvement.

Example predictions
- Most frequent building
- Most frequent floor
- Mean coordinates (x, y) from training data

Key takeaway: The models offer **different trade-offs** in similarity-based prediction, nonlinear learning, randomness, and boosting. Their actual performance must be judged using the **same held-out evaluation data**.

Model Performance on the Official Validation Set

Master model comparison — official validation set (N = 1,111)

Model	Building accuracy	Floor accuracy	Mean error	Median error	P90 error	R^2
Random Forest	**99.91%**	90.19%	**12.28 m**	8.00 m	29.40 m	**0.9791**
Extra Trees	**99.91%**	90.19%	12.63 m	**7.70 m**	**27.00 m**	0.9729
Weighted KNN (k=5)	98.65%	75.79%	14.26 m	8.11 m	27.98 m	0.9532
XGBoost	99.82%	89.11%	22.75 m	15.08 m	49.10 m	0.9425
Dummy baseline	24.12%	15.48%	161.37 m	142.88 m	263.79 m	−0.6412

Evaluation metrics

Accuracy
Percentage of correct building/floor predictions.

F1-score
Balances precision and recall.

Mean error
Average Euclidean coordinate error (in meters).

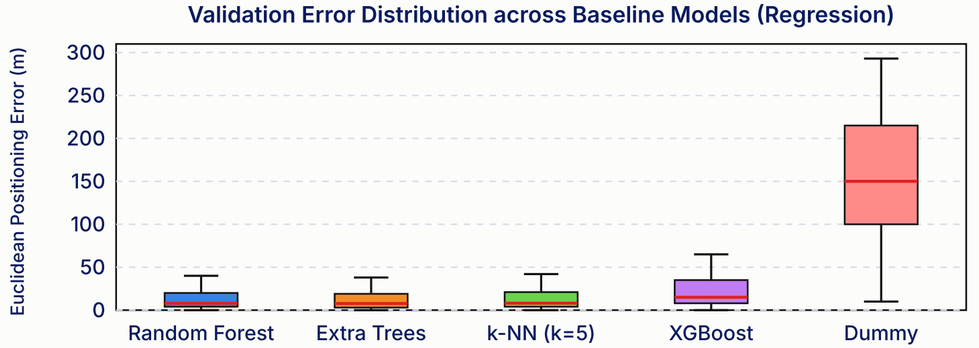
Median error
Middle positioning error (in meters).

P90 error
90th-percentile error (in meters).

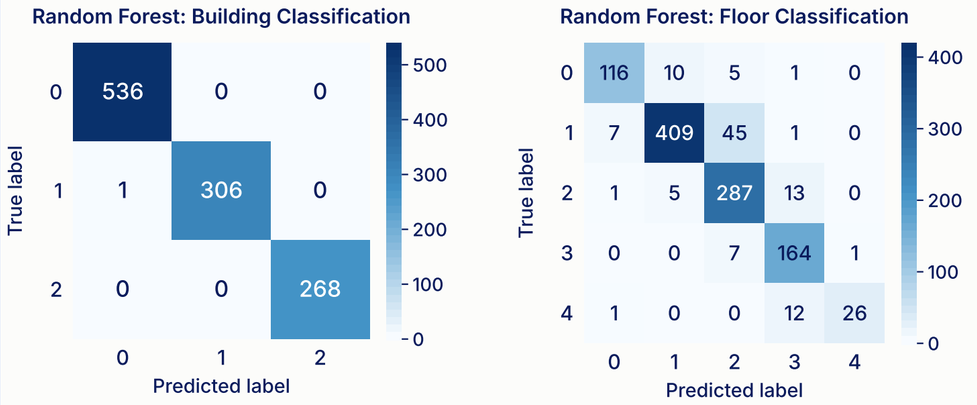
MAE / RMSE
Coordinate-error measures, with RMSE penalizing large errors more.

R^2
How well predicted coordinates explain the variation in actual coordinates.

Validation error distribution (regression)



Random Forest confusion matrices



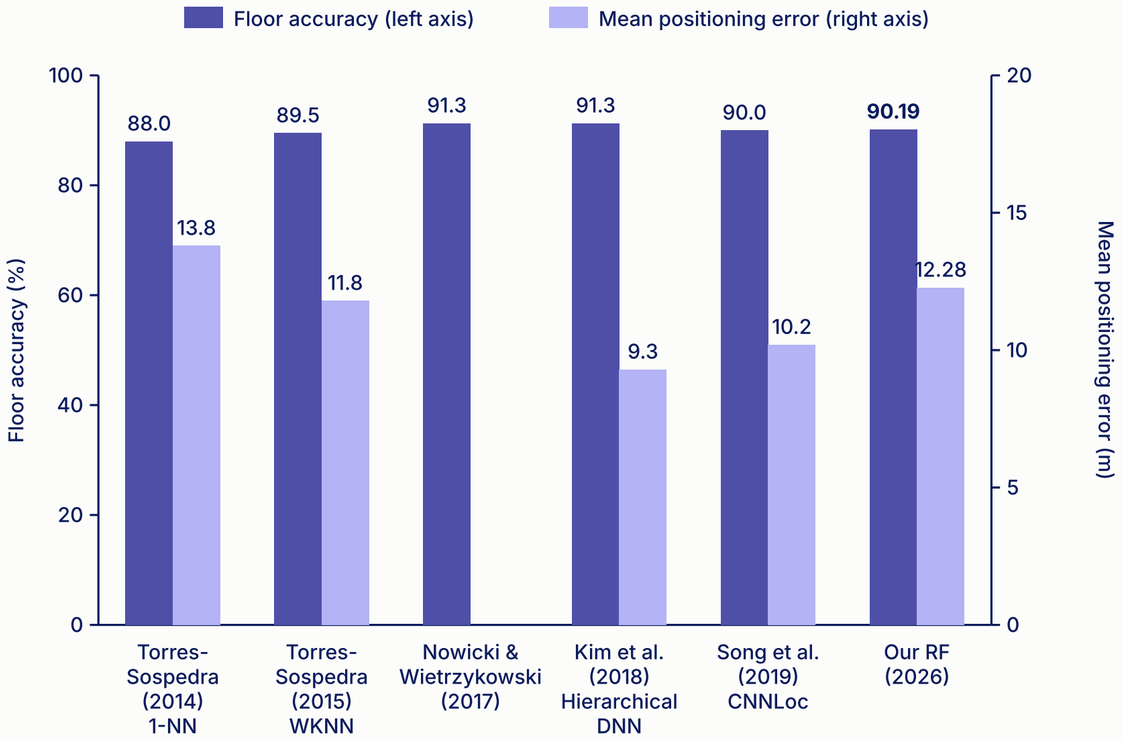
Key interpretation: Random Forest achieved the **lowest mean positioning error** and **highest reported coordinate R^2. Extra Trees** achieved the **lowest median and P90 errors,** so it performed better on these measures of typical and upper-tail error.

Literature Comparison

Comparison with existing approaches

Method	Year	Building accuracy	Floor accuracy	Mean positioning error
Torres-Sospedra et al., 1-NN	2014	100.0%	~ 88.0%	~ 13.8 m
Torres-Sospedra et al., WKNN	2015	100.0%	~ 89.5%	~ 11.8 m
Nowicki & Wietrzykowski	2017	99.8%	91.3%	Not reported
Kim et al., hierarchical DNN	2018	99.8%	91.3%	9.3 m
Song et al., CNNLoc	2019	100.0%	~ 90.0%	~ 10.2 m
Our Random Forest	**2026**	**99.91%**	**90.19%**	**12.28 m**

Floor accuracy and mean positioning error



Key takeaway: Our Random Forest achieves a **competitive floor accuracy (90.19%)** compared to state-of-the-art methods, with a **mean positioning error of 12.28 m**, demonstrating performance comparable to existing approaches.

PROJECT 9

Thank You

Indoor Location Prediction Using Wi-Fi Signals

